

Name	Class	Index No.
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RAFFLES JUNIOR COLLEGE

Preliminary Examination

H1 BIOLOGY

PAPER 2

8875/2

9th September 2008

2 hours

Additional materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group & name on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Write your answers **ONLY in the spaces** provided on the question paper.

Section B

Answer any **ONE** question.

Write your answer on the separate answer paper provided.

All workings for numerical answers must be shown.

At the end of the examination, **hand in your essay SEPARATELY.**

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	X
1	
2	
3	
4	
Section B	
Total	

This document consists of **10** printed pages.

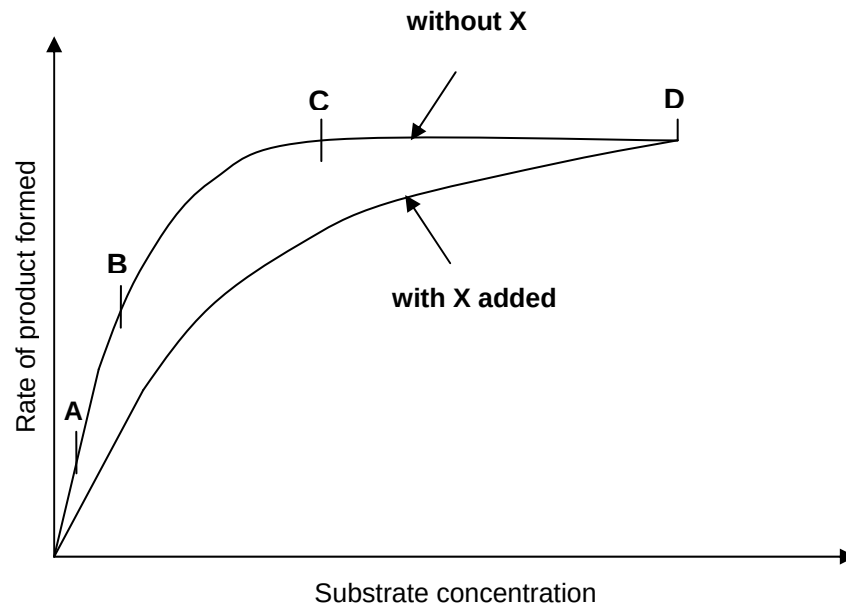


Raffles Junior College
Internal Examination

[Turn over

Section AAnswer **all** the questions in this section.For
Examiner's
Use**Question 1**

- (a) A student carried out an investigation to determine the effect of different concentrations of a substrate on the activity of an enzyme. Another set of experiments was carried out under the same conditions but with a compound **X** added. The results for both experiments were sketched out by the student in Fig.1.1 below.

Fig. 1.1Account for the shape of the graph (without **X**),

- (i) from **A** to **B**, and

.....

[2]

- (ii) from **C** to **D**.

.....

[2]

- (iii) Account for the effect of **X** on the enzyme.

.....

[3]

- (b) When tadpoles turn into frogs, their tails become shorter and finally disappear. Fig. 1.2 shows how the concentration of a protein-digesting enzyme, cathepsin, found in lysosomes changes as a tadpole matures into a frog.

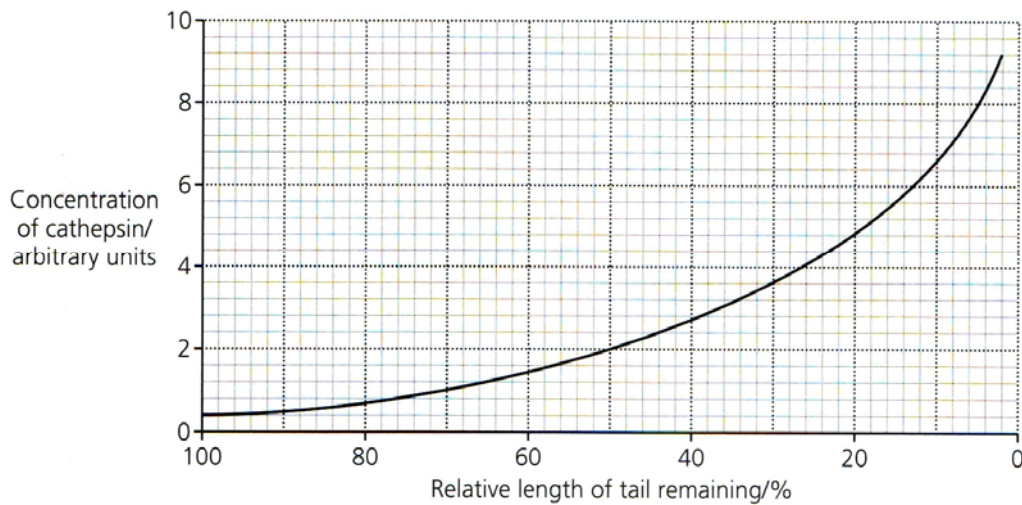


Fig. 1.2

- (i) Suggest two possible causes for the increase in cathepsin concentration in Fig. 1.2.

.....

[2]

- (ii) Explain an advantage to the developing frog when the proteins in the tadpole's tail break down and the products are absorbed.

.....

[2]

- (iii) Apart from protein-digesting enzymes, suggest one other type of digestive enzyme that you would expect to show a similar change in concentration as the tadpole's tail shortens. Give a reason for your answer.

.....

[2]

[Total: 13]

Question 2

Fig.2.1 is a pedigree which traces the inheritance of alkaptonuria, a biochemical disorder. Affected individuals, indicated here by the coloured circles and squares, are unable to break down a substance called alkapton, which colours the urine and stains body tissues.

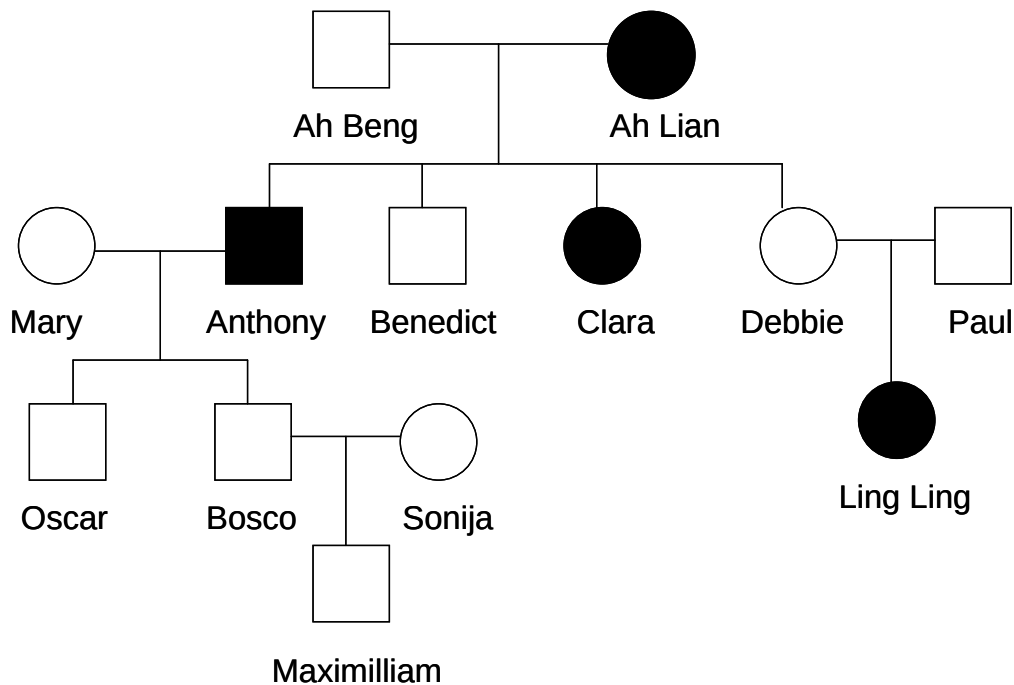


Fig. 2.1

- (a) (i) Is alkaptonuria a dominant or recessive disorder? Explain your answers.

.....
[1]

- (ii) Name **two** individuals whose specific genotypes cannot be determined.

.....
[2]

- (iii) Oscar's genotype is Aa. If he marries a phenotypically normal female, what is the probability that his child will have alkaptonuria? Explain your answer.

.....

[3]

- (b) An inbred variety of maize, variety **A**, with finely striped leaves was found to be resistant to the fungus that causes the disease corn leaf blight.

Plants of variety **A** were crossed with another inbred variety of maize, variety **B**, with entirely green leaves and low resistance to the fungus. All the F_1 generation had entirely green leaves and low resistance.

- (i) Using appropriate symbols (indicating what they represent), state the genotype of **A**.

.....

.....[1]

- (ii) The above F_1 generation was back-crossed to variety **A**. Using a genetic diagram, show the expected phenotype of the F_2 generation from this cross. [3]

[Total: 10]

[Turn over

Question 3

For
Examiner's
Use

- (a) Fig. 3.1 shows cells from an onion root tip. The root tip has been squashed and stained to show the stages of mitosis.

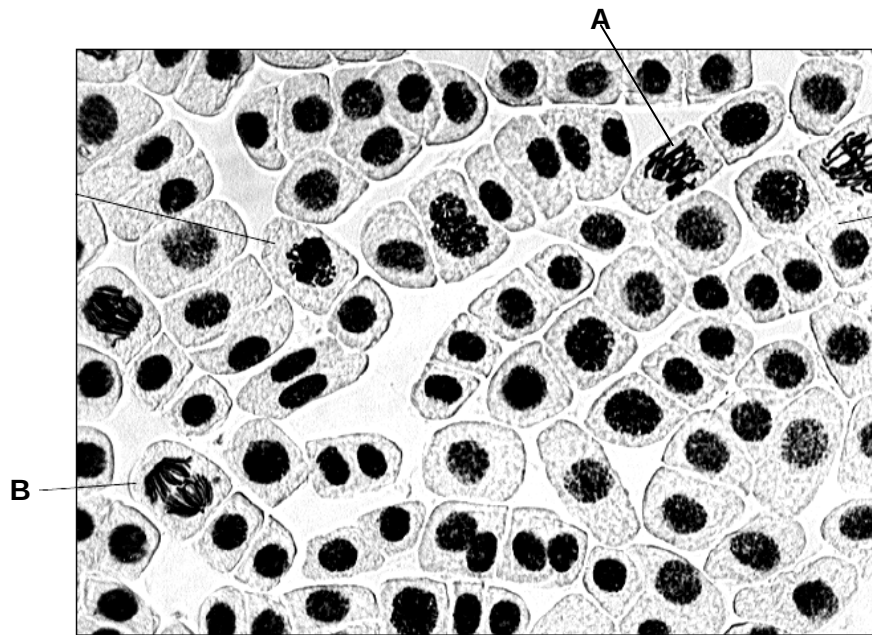


Fig. 3.1

- (i) Describe what happens to the chromosomes during stage **A**.

.....
.....
.....[2]

- (ii) Explain how stage **B** ensures that the daughter cells formed will be genetically identical.

.....
.....
.....[2]

- (iii) The amount of DNA in cells from the onion root tip tissue was analysed. Some cells were found to have 7.6 units of DNA, others had only 3.8 units. Explain why this is so.

.....
.....
.....[2]

[Total: 6]

Question 4

The bacterium *Bacillus thuringiensis* (Bt) makes the Bt-toxin which is toxic to *Lepidopteran* caterpillars and has no effect on most other groups of insects, invertebrates and vertebrates (including Man). The dried bacteria have been used as a biological control measure, sprayed in suspension onto crop plants. However, it becomes ineffective after a few days. The ideal solution would be to have the crop plants make their Bt-toxin. Since Bt-toxin is a protein, this is quite feasible using genetic engineering techniques. Fig.4.1 below shows how tomato plants are genetically engineered to produce their own toxin.

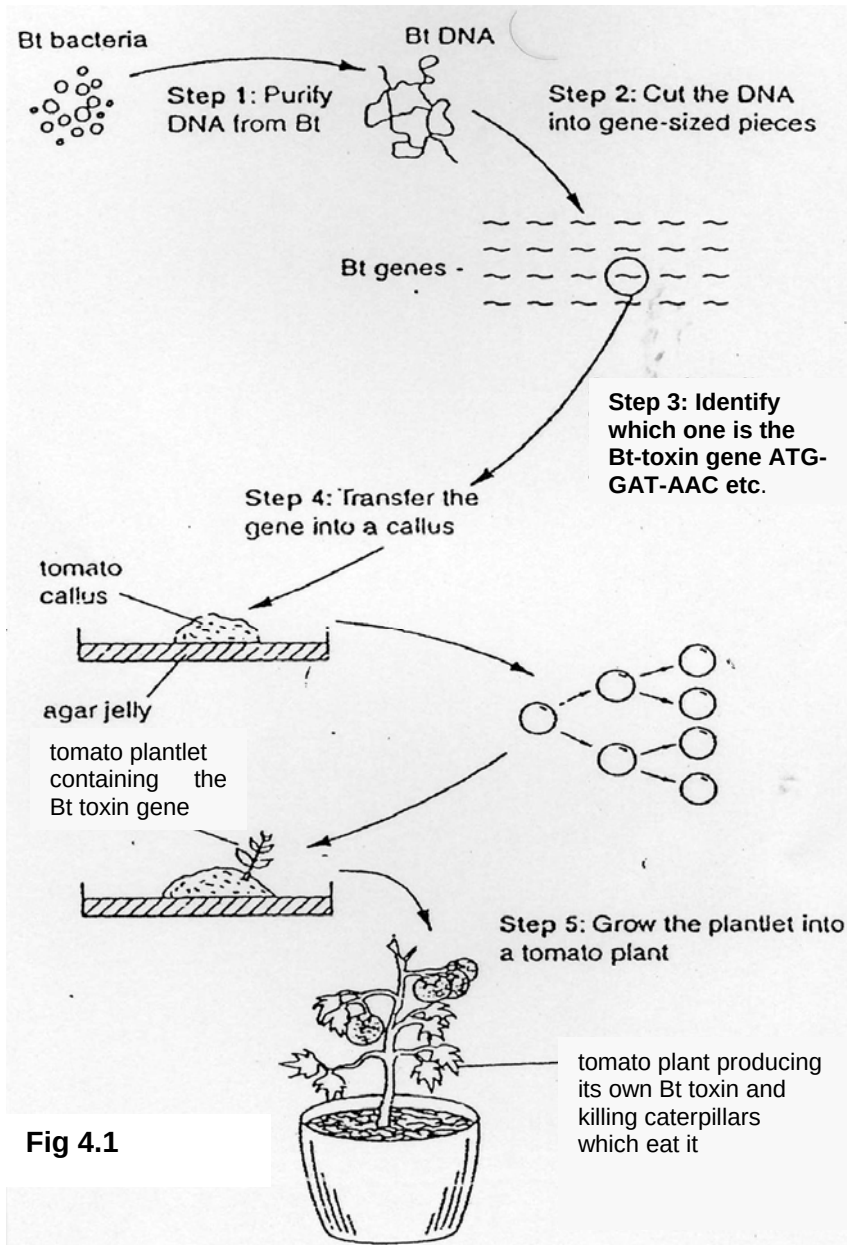


Fig 4.1

(a)(i) State why it was considered a better option to engineer crop plants to make the toxin, rather than to spray the plants with the dried bacteria.

.....

.....[1]

[Turn over

(ii) In **Step 1**, the DNA from *Bt* is extracted and purified. State **two** differences between *Bt* DNA and tomato plant DNA.

1.

.....

2.

.....[2]

(iii) Suggest an alternative method of obtaining the T-toxin gene, instead of the method outlined in the diagram.

.....

.....[1]

(iv) How could the gene be 'tagged' to identify which cells had taken up the gene?

.....

.....[1]

(v) State **two** possible hazards of genetic engineering of crop plants.

1.

.....

2.

.....[2]

- (b) Many crops have been genetically modified (GM) to increase their tolerance to a specific herbicide. Herbicides are chemicals that kill plants.

Britain delayed approving any commercial growing of GM crops until the results of research were available. Research between 2000 and 2003 investigated the effects of GM crops on the environment. Fields were divided and half sown with GM and half with conventional crops. Table shows some of the results of this research.

	Weeds biomass / grams per m ²		Plant-eating insects / mean number per sample		Insect-eating insects / mean number per sample	
	Conventional	GM	Conventional	GM	Conventional	GM
maize	8	14	11.57	15.18	11.09	11.32
beet	22	4	17.14	12.27	15.93	14.73
Oilseed rape	46	13	20.75	16.67	14.15	11.88

Data from *Philosophical Transactions of the Royal Society* 2003

Researchers found different amounts of weeds and insects with different crops.

- (i) With which **conventional** crop did they find the **least** weeds and insects? Justify your answer.

.....
[1]

- (ii) Which GM crop appears to be least harmful to weeds and insects **relative to** the same conventional crop? Justify your answer.

.....

[2]

- (iii) Suggest a reason why the insect population fell when herbicide tolerant GM beet was grown.

.....
[1]

[Total: 11]

[Turn over

Section BAnswer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

Question 5

- (a) You have found the coding sequence of a gene coding for a growth hormone. Outline one method by which this gene could have been

(i) isolated and [4]

(ii) cloned into *E. coli* using a plasmid [8]

for the large scale production of the growth hormone.

(Note: Use a plasmid which carries an ampicillin resistance gene and a tetracycline resistance gene.)

- (b) Describe the role of ribonucleic acids in the Central Dogma. [8]

Question 6

- (a) The products of photosynthesis are used for the formation of storage and structural compounds in plants.

(i) Describe the process by which these storage and structural compounds are formed. [6]

(ii) Discuss the significance of the structure in the functioning of these storage and structural compounds. [7]

- (b) Discuss the 2 categories of transport proteins found in cell membrane. [7]